Choose Appropriate Measures

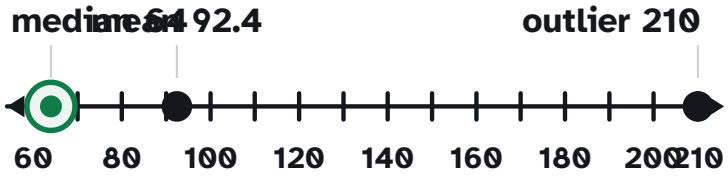
Match the mean with the MAD, or the median with the IQR. Look at the shape first.

Name _____ **Date** _____

Tonight's spark

Two stores list five used-bike prices in dollars. Store 1: 60, 62, 64, 66, 68. Store 2: 60, 62, 64, 66, 210. Both want to post one typical price. Which number is fair for Store 2?

Guess first: is Store 2's mean closer to $64 or to $90?



dollars, Store 2

Our class model: matching center and spread

Try it

Work it out on paper. Write one answer on each line.

1 **A symmetric data set with no outlier is best described by which pair?**

A. Mean and MAD **B.** Median and IQR **C.** Mode and range **D.** Mean and IQR

My answer _____

2 **Which spread measure goes with the median?**

A. The MAD **B.** The IQR **C.** The mean **D.** The outlier

My answer _____

3 **The data 30, 32, 34, 36, 38 are symmetric with no outlier. Which pair should you report?**

A. Median and IQR **B.** Range only **C.** Mean and MAD **D.** Greatest value and least value

My answer _____

4 **Which data set should be described with the median and the IQR?**

A. 10, 11, 12, 13, 14 **B.** 20, 22, 24, 26, 28 **C.** 7, 8, 9, 10, 11 **D.** 3, 4, 4, 5, 60

My answer _____

5 **The data 5, 6, 7, 8, 90 has an outlier. Which pair should you report?**

A. Median and IQR **B.** Mean and MAD **C.** Mean and IQR **D.** Median and MAD

My answer _____

6 **Why use the median and IQR when a data set has an outlier?**

A. They are easier to calculate **B.** They use every value, including the outlier

C. An outlier does not pull them very far, so they still show the typical group

D. The mean cannot be calculated with an outlier

My answer _____

Talk about it

How did the outlier change the mean but not the median?

Home mission

Write five small numbers on sticky notes, like the ages of cousins. Add one huge note, like 100. Put the notes in order. Ask which is fairer to report: the middle note or the mean.

If we get stuck, ask

- Before we compute — is there a value far from all the others?
- Would the mean here describe a typical value, or get dragged?
- Which measure of spread goes with the center you chose?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/2-10-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it